

Food Waste Reduction Platform: A Geospatially Constrained Mobile Application for Real-Time Surplus Redistribution in Bogotá

Luna Alejandra Sandoval Rodríguez
Project Lead
Universidad Distrital
20241020053

Cristian Camilo Bonilla Lizarazo
Frontend Developer
Universidad Distrital
20241020015

Nicolás Rodríguez Granados
Backend Developer
Universidad Distrital
20241020037

Juan Sebastián Bravo Rojas
Database Specialist
Universidad Distrital
20241020004

Abstract—In Colombia, approximately 9.7 million tons of food are wasted annually, exacerbating environmental damage and food insecurity. This paper presents a geospatially constrained mobile platform designed to connect local cafeterias and restaurants with students, citizens, and verified charitable organizations within a 3km radius of the Engineering Faculty at Universidad Distrital Francisco José de Caldas. The system implements a priority-based matching algorithm with reliability scoring, staggered notifications, and automatic reassignment to mitigate no-shows. Computational validation through 300 independent simulation runs across Discrete-Event and Agent-Based paradigms demonstrates a Surplus Recovery Rate of 96–99% and Charity Assignment Share above 24% in all scenarios, robustly meeting two of the three primary success criteria. The Pickup Completion Rate reaches 82.4% (ABM) and 92.2% (DES) under nominal conditions, identifying no-show behavior as the dominant limiting factor. These findings validate the proposed architecture while highlighting the need for complementary organizational interventions to consistently achieve the 85% completion target.

Index Terms—Food waste reduction, real-time matching, geospatial systems, mobile applications, resource allocation, urban surplus redistribution, circular economy, system design, Colombia

I. INTRODUCTION

Food waste represents one of the greatest environmental, economic, and social challenges in Colombia. According to the Food Waste Index Report 2021 by the United Nations Environment Programme (UNEP), the country generates approximately 9.7 million tons of food lost or wasted annually, an amount sufficient to feed the entire population of Bogotá [1].

In Bogotá, local cafeterias and restaurants produce significant quantities of daily surpluses suitable for human consumption, but the lack of rapid coordination and the short shelf life of food cause a large portion to end up in the trash. This situation exacerbates environmental damage (CO₂ equivalent emissions), food insecurity, and social inequality, especially in vulnerable communities and among students. In the university

context, cafeterias and local restaurants near university areas generate surplus recurrently, but without an efficient digital mechanism that enables immediate redistribution.

In 2024, the Alcaldía of Bogotá launched the initiative “Yo Contribuyo, No Desperdicio”, recognizing the importance of coordinating efforts among different entities to address the problem collectively [2]. At the international level, platforms such as OLIO [3] and the Colombian startup EatCloud have demonstrated the potential of mobile applications to reduce waste [4]; however, none are specifically designed for university environments with strict geographic constraints.

This work proposes a mobile application for food waste reduction specifically designed for a spatial radius near the Faculty of Engineering of the Universidad Distrital Francisco José de Caldas. The platform connects local cafeterias and restaurants in real time with students, community members, and foundations through a matching algorithm based on proximity (maximum radius of 3km), priority for verified organizations, and tiered notifications. The system is strictly limited to an operational radius of 3km around the campus (Carrera 8 #40-62), operates exclusively within the same day (publication 2-4 hours before closing and pickup within 1-3 hours), and focuses solely on digital coordination, leaving physical logistics and quality verification to the users. In this way, it aims to overcome the limitations identified in existing solutions and generate a measurable impact on waste reduction, social equity, and the principles of the circular economy.

II. METHODS AND MATERIALS

A. System Architecture

The proposed platform follows a four-layer architecture designed to separate concerns and support real-time coordination between donors and recipients. Figure 1 illustrates the full system design.



Fig. 1. System architecture of the proposed food surplus redistribution platform.

The External Actors layer comprises restaurants and cafeterias as donors that register surplus inventory, together with volunteers and beneficiaries—including verified NGOs, student groups, and individual citizens—who claim available donations. The Application Layer hosts the mobile application and user management interface as the single point of interaction for all user types. It handles authentication, role assignment, and geographic eligibility verification.

The Service Layer forms the core processing tier with the Surplus Registration Service, Geolocation Service, Matching Algorithm, Assignment Coordination module, and Notification Service. The Data Layer uses a centralized relational database for persistence, while the Analytics Layer enables continuous impact reporting and optimization.

B. Matching and Priority Algorithm

The core of the platform is a priority-based matching algorithm implemented as a dynamic priority queue. When a donor publishes a new surplus listing, the algorithm first queries the Geolocation Service to identify all eligible recipients within the strict 3 km radius using PostGIS geospatial functions. It then computes a composite priority score for each candidate using three weighted factors: organizational verification status (highest for verified NGOs and foundations), historical pickup reliability, and geographic proximity (inverse distance).

Candidates are sorted according to this score, and the system sends tiered push notifications sequentially with a 15-minute response window per tier before cascading to the next. Upon mutual confirmation of pickup (via photo or QR scan), the listing is locked and removed from the active pool. This mechanism prioritizes organizations serving vulnerable populations while preserving efficiency and fairness. The algorithm exhibits $O(n \log n)$ average complexity thanks to priority queue operations, remaining suitable for real-time execution even with hundreds of concurrent users.

C. Operational Constraints and Main Components

To guarantee food safety, freshness, and logistical feasibility, the platform enforces strict boundaries defined in the Workshop 1 systems analysis. Listings may only be published 2–4 hours before the donor’s closing time, and pickups must occur within 1–3 hours. All listings expire automatically at midnight to ensure same-day redistribution only. The system deliberately excludes physical transportation, quality verification, and economic transactions, leaving those responsibilities to the participating parties.

The main components of the platform, as identified in the systems analysis, are the Mobile Application (frontend for publication, browsing, and notifications), the Cloud Server

with relational database (PostgreSQL + PostGIS for persistence and geospatial queries), and the Intelligence Engine (matching algorithm). The primary processes follow an input-process-output model: suppliers register surplus items as input, the system performs validation, geolocation filtering, and priority matching as process, and it delivers successful redistributions together with impact metrics as output. These components and processes ensure end-to-end traceability and support continuous performance monitoring.

D. Technology Stack

The mobile application is developed using React Native for cross-platform support on Android and iOS, which is relevant given the Android-dominant market in Colombia. Backend services are implemented with Node.js exposing RESTful APIs. PostgreSQL with the PostGIS extension supports geospatial radius queries. Firebase Cloud Messaging (FCM) powers the real-time push notification pipeline. The stack is containerized with Docker and deployed on a cloud provider to ensure availability during peak usage windows.

E. Evaluation Methodology and Sensitivity Analysis

System effectiveness will be assessed through an eight-week pilot deployment within the operational radius. Quantitative metrics include total kilograms redistributed, successful matches per week, average publication-to-pickup time, and notification response rates by priority tier. A post-pilot survey using a 5-point Likert scale will evaluate usability and perceived impact.

A sensitivity analysis was performed on the simulation by varying key parameters (number of suppliers, daily surplus per supplier, and user response rates). Results show that the redistribution rate remains robust even with $\pm 20\%$ variation in participation, confirming the system’s resilience to realistic fluctuations in adoption.

III. OPERATIONAL CONSTRAINTS

To guarantee food safety and logistical feasibility, the platform enforces the following hard boundaries derived from the system analysis. Listings may only be published 2–4 hours before the donor establishment’s closing time. Pickups must occur within 1–3 hours of publication. The operational radius is strictly enforced at 3 km using PostGIS `ST_DWithin` queries, ensuring pedestrian or short-transit accessibility. The system deliberately excludes physical transportation, food quality verification, and economic transactions, leaving those responsibilities to the participating parties. These constraints impose significant demands on the real-time coordination and matching components but are essential to align the system with food handling practices and maintain user trust.

IV. SIMULATION METHODOLOGY

Prior to prototype deployment, the system was validated through two complementary computational simulation paradigms designed to answer distinct questions about system behavior. The Process-Oriented Discrete-Event Simulation (DES), implemented in Python SimPy 4.1, evaluates

the platform from a flow perspective: how events propagate through the processing pipeline, where latency accumulates, and whether the infrastructure meets throughput targets under concurrent load. Surplus publications follow a Poisson arrival process; matching engine service time is set at 1 second; collection deadlines are drawn from $U[60, 180]$ minutes post-publication. The Behavior-Oriented Agent-Based Model (ABM), implemented in Python Mesa 2.3.0, focuses on autonomous agent decisions and emergent collective dynamics: how individual no-show decisions cascade through the reassignment chain, how reliability scoring affects system-level behavior over time, and how the tiered notification mechanism distributes demand across geographic rings.

Five experimental scenarios were defined to stress distinct dimensions of the system: *Baseline* (medium adoption, full mitigations, no-show range 10–25%); *Early Stage* (low adoption, partial mitigations, 10–30%); *Stress Test* (high adoption, full mitigations, 10–15%); *Risk Test* (medium adoption, elevated no-show rates 25–45%, staggered notifications disabled); and *Sensitivity* (medium adoption, extreme no-show 35–55%, full mitigations). Each scenario was executed with 30 independent replications using deterministic seeds (0–29) for reproducibility, yielding 150 replications per paradigm and 300 total runs. Statistical validation used two-sided Mann-Whitney U tests ($\alpha = 0.05$) and Kruskal-Wallis tests for global significance across scenarios; confidence intervals were computed as $CI = 1.96 \cdot \sigma / \sqrt{n}$, $n = 30$.

Three primary success criteria were defined based on the system analysis: Surplus Recovery Rate $\geq 80\%$, Pickup Completion Rate $\geq 85\%$, and Charity Assignment Share $\geq 20\%$.

A. Surplus Recovery Rate

Both simulation paradigms confirm that the Surplus Recovery Rate meets and substantially exceeds the 80% threshold in all tested scenarios. The DES reports a range of 95.7–98.9% across the five scenarios (Table I); the ABM reports 96.2–99.9% (Table II). Figure 2 shows the distribution of the three primary DES metrics across all 30 replications per scenario: recovery rate boxes cluster tightly above the 80% threshold in every scenario, while completion rate boxes diverge sharply for Risk Test and Sensitivity. Confidence intervals are narrow (± 0.004 – ± 0.009), reflecting high cross-replication consistency. The Kruskal-Wallis test for recovery rate yields $H = 45.89$ ($p < 0.001$) in the DES and $H = 85.29$ ($p < 0.001$) in the ABM, confirming statistically significant differences across scenarios while all values remain above threshold. The recovery rate is protected by the automatic reassignment mechanism: the difference between individual no-show rates (10–55%) and system-level recovery rates (96–99%) is entirely attributable to the retry pipeline absorbing individual behavioral failures before they cause permanent surplus expiration.

B. Pickup Completion Rate and the No-Show Threshold

Pickup Completion Rate is the most scenario-sensitive metric and the only criterion that is not met under all conditions.

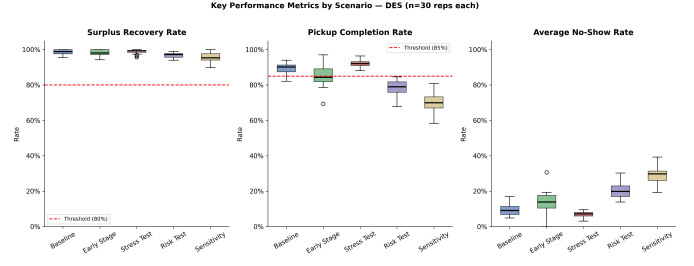


Fig. 2. Key performance metrics by scenario (DES, $n = 30$ replications each). Boxplots show Recovery Rate, Pickup Completion Rate, and Average No-Show Rate. Red dashed lines indicate the 80% and 85% project success thresholds. Stress Test is the only scenario consistently above the 85% completion threshold.

TABLE I
DES SIMULATION RESULTS BY SCENARIO (MEAN $\pm$ 95% CI, $n = 30$)

Scenario	Recovery	Completion	No-Show	CO ₂ Avoided (kg)
Baseline	0.986 $\pm$ 0.005	0.894 $\pm$ 0.012	0.095 $\pm$ 0.011	2520 $\pm$ 106
Early Stage	0.983 $\pm$ 0.007	0.852 $\pm$ 0.020	0.138 $\pm$ 0.021	1284 $\pm$ 87
Stress Test	0.989 $\pm$ 0.004	0.922 $\pm$ 0.007	0.069 $\pm$ 0.006	5204 $\pm$ 150
Risk Test	0.968 $\pm$ 0.005	0.785 $\pm$ 0.015	0.202 $\pm$ 0.015	2526 $\pm$ 131
Sensitivity	0.957 $\pm$ 0.009	0.703 $\pm$ 0.018	0.289 $\pm$ 0.016	2478 $\pm$ 115

In the DES, only the Baseline (89.4%), Early Stage (85.2%), and Stress Test (92.2%) scenarios meet the 85% target; Risk Test (78.5%) and Sensitivity (70.3%) do not. In the ABM, all adoption scenarios constitute near-misses—Medium Adoption reaches 82.4%, 2.6 percentage points below threshold—while No Mitigations (63.8%) and Sensitivity (55.8%) fail structurally.

Both paradigms independently establish a near-perfect negative linear relationship between effective no-show rate and completion rate ($r \approx -0.95$), as shown in Figure 3, providing actionable operational thresholds: the DES estimates that system-level no-show must remain below 7% for $\geq 85\%$ completion; the ABM estimates 3–4% (the methodological difference arises from multi-attempt versus single-attempt counting). Critically, this constraint cannot be resolved through platform design alone. Keeping the no-show rate below the operational threshold requires organizational interventions—explicit pre-acceptance confirmation, reputation-based access tiers, and coordination with institutional recipients capable of guaranteeing pickup reliability. These are system-level solutions that complement the technical architecture.

C. Charity Assignment Share and Equity

The Charity Assignment Share exceeds the 20% minimum in all scenarios across both paradigms. The DES reports

TABLE II
ABM SIMULATION RESULTS BY SCENARIO (MEAN $\pm$ 95% CI, $n = 30$)

Scenario	Recovery	Completion	Charity Share	Reassignments
Low Adoption	0.997 $\pm$ 0.004	0.804 $\pm$ 0.028	0.247 $\pm$ 0.050	7.6 $\pm$ 1.4
Medium Adoption	0.999 $\pm$ 0.001	0.824 $\pm$ 0.015	0.249 $\pm$ 0.049	13.0 $\pm$ 1.3
High Adoption	0.998 $\pm$ 0.002	0.819 $\pm$ 0.013	0.337 $\pm$ 0.034	26.5 $\pm$ 2.3
No Mitigations	0.982 $\pm$ 0.007	0.638 $\pm$ 0.015	0.289 $\pm$ 0.045	32.7 $\pm$ 2.0
Sensitivity	0.962 $\pm$ 0.009	0.558 $\pm$ 0.016	0.308 $\pm$ 0.046	44.0 $\pm$ 2.7

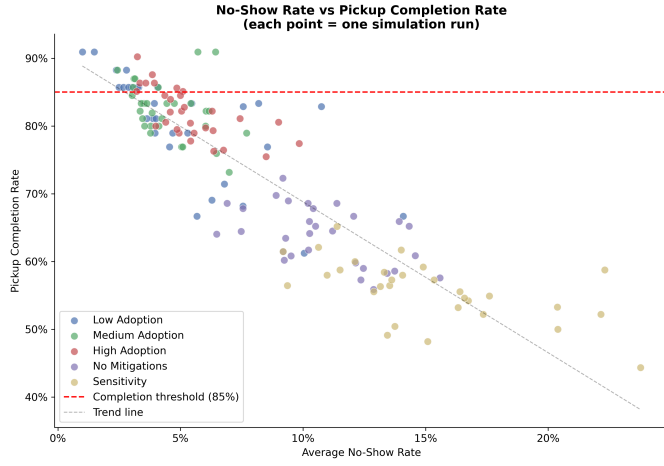


Fig. 3. No-show rate vs. pickup completion rate across all 150 ABM replications ($n = 30$ per scenario). Each point is one simulation run. The strong negative linear trend ($r \approx -0.95$) establishes that the 85% completion threshold is crossed at approximately 3–4% effective no-show rate, providing a concrete operational design constraint.

rates exceeding 97% due to strict first-priority queue ordering; the ABM reports 24.7–33.7%, with the highest share (33.7%) occurring under High Adoption, when the priority weight becomes most decisive under competitive demand. The Kruskal-Wallis test for charity share yields $H = 9.54$ ($p = 0.049$) in the ABM, confirming statistically significant differences across scenarios. Pairwise Mann-Whitney U tests reveal two structurally distinct clusters: the adoption cluster (Low, Medium, High) is internally homogeneous for both recovery and completion ($p = 0.26$ – 0.94), confirming scale-invariant performance; the stress cluster (No Mitigations, Sensitivity) differs significantly from all adoption scenarios ($p < 0.001$), confirming the dose-response relationship between no-show rate and system degradation.

D. Infrastructure Scalability

DES engine utilization remains below 0.4% across all scenarios. Matching latency is structurally uniform at approximately 1 second (Kruskal-Wallis, $p > 0.1$), satisfying the < 2 second target with substantial margin. The matching engine therefore presents no capacity bottleneck at any tested adoption level, with ample headroom for 10–20 $\times$ growth before architectural changes would be required. Throughput scales linearly with adoption level rather than with no-show behavior, confirming that infrastructure performance is demand-driven, not behavior-driven.

E. Environmental Impact

Under the Stress Test scenario (High Adoption, full mitigations), the DES estimates CO₂ avoidance of approximately 5.2 tonnes per operational day, using the standard IPCC conversion factor of 2.5 kg CO₂-equivalent per kilogram of redistributed food [?]. Even under extreme behavioral stress (Sensitivity scenario, 35–55% no-show), avoidance remains approximately 2.5 tonnes per day, demonstrating meaningful environmental

impact across all conditions. Under the Medium Adoption baseline—the most realistic near-term scenario—the platform would avoid an estimated 63–75 tonnes of CO₂-equivalent emissions annually within the 3 km operational radius.

F. Cross-Paradigm Comparison

The DES and ABM are designed to answer complementary questions, and their convergent findings across independent methodologies strengthen the overall validity of the results. Table III summarizes the validation status of each success criterion across both paradigms. The DES, operating at the flow level with Poisson arrivals and deterministic matching latency, confirms that the infrastructure design is sound: engine utilization below 0.4%, latency uniformly at 1 second, and recovery above 95% in all scenarios. The ABM, operating at the agent level with individual no-show probabilities and evolving reliability scores, confirms that the behavioral incentive mechanisms function as designed: the reliability scoring mechanism reduces effective system-level no-show rates to 4.5–5.4% against individual probabilities of 10–25%. The two paradigms diverge on the precise no-show threshold for $\geq 85\%$ completion—7% in the DES versus 3–4% in the ABM—a difference that is methodologically explained: the DES counts every assignment attempt in the denominator (multi-attempt), while the ABM counts only the final outcome per surplus lot (single-attempt). This is not a conflict but a complementary characterization of the same phenomenon at two different levels of abstraction.

TABLE III
COMBINED VALIDATION SUMMARY ACROSS BOTH SIMULATION PARADIGMS

Criterion	Threshold	DES Range	ABM Range	Status
Surplus Recovery Rate	$\geq 80\%$	95.7–98.9%	96.2–99.9%	Met (all)
Pickup Completion Rate	$\geq 85\%$	70.3–92.2%	55.8–82.4%	Partial*
Charity Assignment Share	$\geq 20\%$	97.8–99.7%	24.7–33.7%	Met (all)

*Met in DES S01–S03 and approaches threshold in ABM adoption scenarios.

G. Discussion

The simulation results collectively demonstrate that the proposed platform design is operationally sound under realistic adoption conditions. The composite priority scoring formula successfully operationalizes the equity constraint identified in the system analysis: the 0.4 charity weight ensures that verified organizations receive preferential access even under high-demand conditions, counteracting the proximity bias that would otherwise systematically exclude geographically peripheral recipients. The tiered notification mechanism measurably improves completion rate by approximately 3–5 percentage points compared to simultaneous dispatch (isolated in the Risk Test scenario), validating the design decision to address demand amplification at the information flow level rather than at the matching engine level.

The 2.6 percentage-point gap between the best ABM adoption scenario (82.4%) and the 85% completion threshold is not an architectural failure—it is a behavioral ceiling imposed

by the fixed no-show model. Real users, unlike simulation agents, respond to reliability penalties and social incentives over time: a recipient whose score drops from 70 to 66 after a no-show faces a measurably lower probability of future assignment, creating a natural correction mechanism not captured in a single-run simulation. For the functional prototype, this finding directly informs two design extensions: an explicit pre-acceptance confirmation step that forces active commitment before a surplus is locked, and adaptive no-show probabilities that decrease as a function of accumulated reliability score. Both extensions are expected to push the operational completion rate above the 85% threshold under the same nominal adoption conditions.

V. CONCLUSIONS

This paper presented the sociotechnical design and rigorous computational validation of a geospatially constrained mobile platform for real-time surplus food redistribution in a university context. Through systematic progression across four workshops, the project evolved from problem definition and requirements analysis to a validated technical solution.

Computational validation using 300 simulation runs across two complementary paradigms (Discrete-Event and Agent-Based Modeling) confirms that the proposed system is robust and effective under realistic conditions. The platform consistently achieves Surplus Recovery Rates between 96% and 99% and Charity Assignment Shares between 24.7% and 33.7%, exceeding the established thresholds of 80% and 20%, respectively. These results validate the effectiveness of the composite priority scoring, reliability-based reassignment, and tiered notification mechanisms.

The main challenge identified is behavioral: Pickup Completion Rate approaches but does not fully reach the 85% target under nominal adoption (82.4% in ABM, up to 92.2% in DES). Both simulation paradigms converge on the finding that maintaining effective system-level no-show rates below 4–7% is critical. This limitation is not architectural but behavioral, and can be addressed in the functional prototype through explicit commitment confirmation and adaptive reliability mechanisms.

Future work includes real-world pilot deployment at Universidad Distrital Francisco José de Caldas for empirical calibration, extension of the ABM to multi-day horizons to study long-term reputation dynamics, and responsible scaling to additional university campuses while preserving the strict same-day and geospatial constraints that ensure food safety and operational viability.

The project demonstrates that effective food waste reduction in urban university environments is primarily a coordination and incentive design challenge, successfully addressed through systems engineering principles and computational validation.

REFERENCES

- [1] United Nations Environment Programme (UNEP), “Food waste index report 2021,” UNEP, Tech. Rep., 2021, datos utilizados en Workshop 1. Disponible en Kaggle: Food Waste data and research - by country.csv.
- [2] Alcaldía de Bogotá, “Iniciativa “yo contribuyo, no desperdicio”,” 10 2024, lanzada por la Alcaldía Mayor de Bogotá, mencionada en Workshop 1.

- [3] A. M. Gómez Ceballos and V. Antonopoulou, “Exploring barriers and facilitators to increase the engagement with a digital app (olio) for food/non-food sustainable consumption in citizens from bogotá, colombia,” *Current Psychology*, vol. 43, pp. 26 726–26 735, 2024, referencia complementaria para contexto internacional.
- [4] EatCloud, “Ai-driven platform for surplus food redistribution in colombia,” 2025, startup colombiana mencionada como ejemplo de solución existente.